

CLOUD COMPUTING LABORATORY

CSA1511-CO3-AT1-EXPERIMENTS

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EXPERIMENT 1: Library Book Reservation System – SaaS

AIM

To create and deploy a simple cloud-based Library Book Reservation System using any Cloud Service Provider to demonstrate Software as a Service (SaaS).

PROCEDURE

1. Select a Cloud Service Provider such as Firebase, Google Cloud, AWS, Azure, or any suitable cloud platform.
2. Create a simple web-based Library Book Reservation application.
3. Create a database/table to store library book details.
4. Add the required fields: Title of the Book, Author, Publication, Year, Edition, Number of Copies, Rack Number, and Status.
5. Provide options to add, search, reserve, and update books.
6. Set the book status as Taken when a book is reserved/issued and Returned when it is returned.
7. Connect the application to the cloud database and store the book records online.
8. Deploy the application using the selected cloud service so that it can be accessed through a web browser.
9. Test the application by adding sample books and updating their reservation status.

INPUT

- Book Title: Operating System Concepts
- Author: Abraham Silberschatz
- Publication: Wiley
- Year: 2023
- Edition: 10th Edition
- No. of Copies: 5
- Rack No.: A-12
- Status: Returned

RESULT

EXPERIMENT 2: Payroll Processing System – SaaS

AIM

To create and deploy a simple cloud-based Payroll Processing System using any Cloud Service Provider to demonstrate Software as a Service (SaaS).

PROCEDURE

10. Select a Cloud Service Provider and create a simple web-based Payroll Processing application.
11. Create a database/table for storing employee payroll information.
12. Add the required fields: Employee Name, Experience in Present Organization, Overall Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
13. Enter employee details and salary components into the application.
14. Calculate Gross Pay using the salary components.
15. Calculate Net Pay after considering applicable deductions, if any.
16. Store the employee payroll records in the cloud database.
17. Deploy the application on the selected cloud platform.
18. Test the application by entering sample employee records and verifying the calculated salary values.

INPUT

- Employee Name: Rahul
- Experience in Present Organization: 2 Years
- Overall Experience: 4 Years
- Basic Pay: ₹30,000
- HRA: ₹6,000
- DA: ₹3,000
- TA: ₹2,000
- Deductions: ₹1,000
- Gross Pay: ₹41,000
- Net Pay: ₹40,000

RESULT

Payroll Processing System

Trial expires in 3 days Upgrade Edit this a

PPS Payroll Processing System

Pay Master

Pay Master

Pay Master Report

Payroll Processing System

MANEDI PRANAY

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My Account

Home

Sign out

Pay Master

Employee ID *
Employee Name *
Department *
Designation
Basic Salary
HRA
DA
Other allowances
Gross Salary
PF Deduction
Tax Deduction
Other Deduction
Net Salary *

Payroll Processing System

Trial expires in 3 days Upgrade Edit this application Help

PPS Payroll Processing System

Pay Master

Pay Master

Pay Master Report

Payroll Processing System

MANEDI PRANAY

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My Account

Home

Sign out

Pay Master Report

Employee ID	Employee Name	Department	Designation	Basic Salary	HRA	DA
EMP008	Rohith	Sales	Java+	9546456	684684	46846
EMP007	Giri	Sales	c+	45965	654	6846
EMP006	Sai	Marketing	c++	94631	65465	6455322
EMP005	Pranay	Sales	Java	4798	8646	854563
EMP004	SREEKAR	IT	AI	35468	554	5451
EMP003	Ariya	HR	Accountant	45000	9000	4500
EMP002	Prince	Finance	HR	5000	1000	500
EMP001	Sai Rahul	HR	Software Engineering	50000	10000	5000

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A cloud-based Payroll Processing System was successfully created and deployed. Employee salary details were stored in the cloud, and gross pay and net pay were calculated successfully, demonstrating SaaS.

EXPERIMENT 3: Type-2 Hypervisor – VM Creation and Configuration

AIM

To demonstrate virtualization by installing a Type-2 Hypervisor and creating a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage using Windows/Linux as the guest operating system.

PROCEDURE

19. Install a Type-2 Hypervisor such as Oracle VirtualBox or VMware Workstation on the host computer.
20. Open the installed hypervisor and select the option to create a new Virtual Machine.
21. Enter a suitable name for the virtual machine and select Windows or Linux as the guest operating system.
22. Allocate 1 CPU/core to the virtual machine.
23. Allocate 2 GB (2048 MB) of RAM to the virtual machine.
24. Create a virtual hard disk with a storage capacity of 15 GB.
25. Attach the required Windows/Linux ISO image to the virtual optical drive.
26. Start the virtual machine and install the selected guest operating system.
27. After installation, boot the VM and verify the allocated CPU, RAM, and storage.
28. Test the VM by opening applications or terminal commands inside the guest operating system.

INPUT

- Host OS: Windows/Linux
- Type-2 Hypervisor: Oracle VirtualBox / VMware Workstation
- Guest OS: Linux/Windows
- CPU: 1 Core
- RAM: 2 GB
- Storage: 15 GB
- ISO Image: Selected Windows/Linux ISO

RESULT

A Type-2 Hypervisor was successfully installed and a virtual machine was created with 1 CPU, 2 GB RAM, and 15 GB storage. The selected guest operating system was installed and executed successfully inside the VM.

Type-2 Hypervisor Virtual Machine Creation

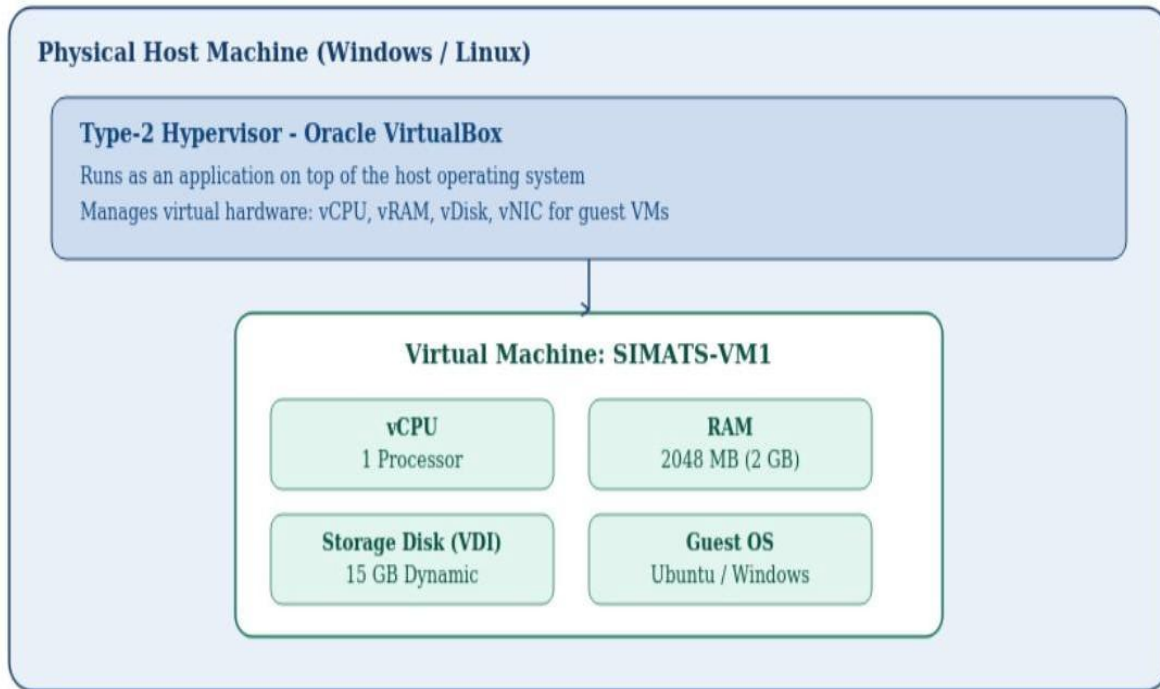


Fig 3.1 — VirtualBox VM configuration: 1 vCPU, 2 GB RAM, 15 GB storage

EXPERIMENT 4: VM Cloning and Loading the Previous/Cloned Version

AIM

To demonstrate virtualization by cloning an existing virtual machine using a Type-2 Hypervisor and testing the cloned VM by loading and running the previous/cloned version.

PROCEDURE

29. Install and open a Type-2 Hypervisor such as Oracle VirtualBox or VMware Workstation.
30. Create or select an existing virtual machine containing the required guest operating system.
31. Start the original VM and create a sample file or make a simple configuration change to identify the original version.
32. Shut down the original VM properly.
33. Select the VM and choose the Clone option from the hypervisor.
34. Provide a suitable name for the cloned VM.
35. Select the appropriate clone type, such as Full Clone, so that the cloned VM has its own virtual disk.
36. Complete the cloning process and wait until the clone is created.
37. Start the cloned VM and verify that the guest operating system loads successfully.
38. Check the sample file/configuration to confirm that the previous VM state has been copied to the clone.
39. Make a change in the cloned VM and verify that the original VM remains unchanged.

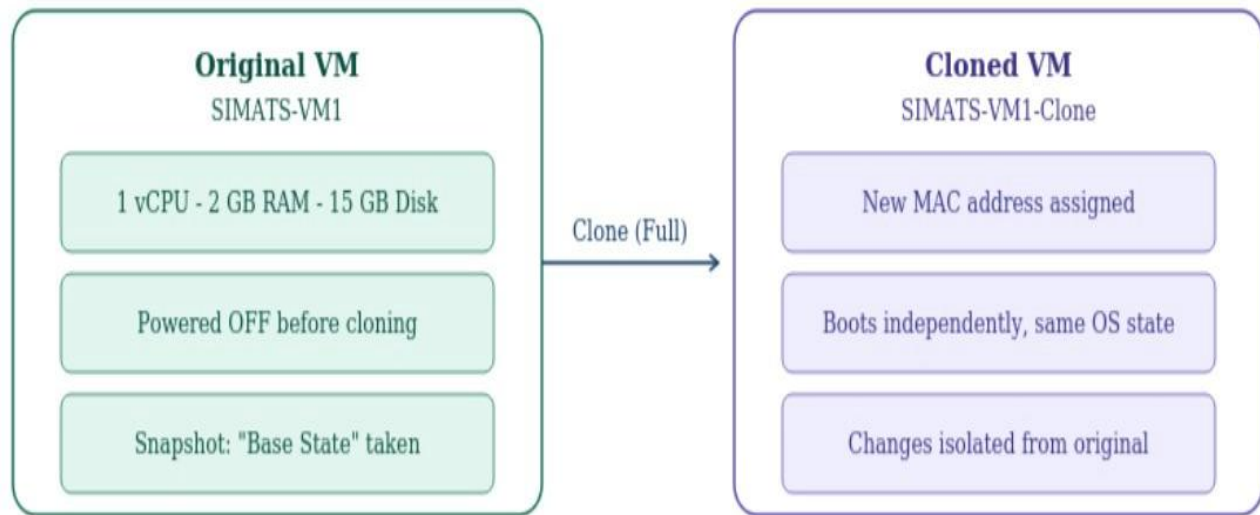
INPUT

- Original VM: Linux/Windows VM
- Hypervisor: Oracle VirtualBox / VMware Workstation
- Clone Type: Full Clone
- Cloned VM Name: Linux-Clone / Windows-Clone
- Test File: sample.txt
- Test Operation: Boot cloned VM and verify previous VM contents

RESULT

The existing virtual machine was successfully cloned using the Type-2 Hypervisor. The cloned VM loaded and operated correctly with the previous VM contents. Changes made to the cloned VM did not affect the original VM, demonstrating VM cloning and virtualization.

VM Cloning and Independent Testing



Test: a file created only in the clone does not appear in the original - confirming independence

Fig 4.1 — Cloning workflow: original VM cloned to an independent instance